6 Results, validation and discussion

Section 5 proposed comparisons across objects, loads and supports. Those experiments are not yet available, so we first examine what the completed square example establishes. We use Section 4's recorded solver results and new square-only refinement and load-sensitivity checks, reusing the same implementation [1]. These results do not represent the proposed additional objects.

6.1 Deformation under mesh refinement

We keep Section 4's material, full left clamp and uniform right-edge traction fixed. Table 1 compares structured meshes with n cells per side, retaining the same diagonal in every cell. These are separate from Section 5's proposed Triangle meshes. Each physical measurement uses a checked sparse-LU solution.

Table 1: New square-only refinement results from `generate_evidence.py`. Here $\bar{u}_x$ is the right-edge horizontal displacement averaged by edge length; u_x^m, u_y^m are the components at the fixed midpoint $(1, 0.5)$ m. Displacements are in mm. $\kappa_2(A)$ is estimated from checked extreme eigenpairs.

n	Free unknowns	$\bar{u}_x$	u_x^m	u_y^m	$\kappa_2(A)$
5	60	0.491352	0.489202	0.010710	6.72×10^2
10	220	0.492877	0.490810	0.003769	2.47×10^3
20	840	0.493497	0.491599	0.001255	9.10×10^3
40	3280	0.493744	0.491900	0.000418	3.45×10^4

The mean extension increases towards about 0.494 mm, while its successive changes shrink from 0.00153 to 0.000620 and 0.000247 mm. The midpoint's vertical movement also decreases. We integrate the edge trace rather than average nodes, and interpolate at the fixed midpoint when it is not a node. Absolute differences avoid dividing by its small vertical component. Every reported mesh difference exceeds the summed algebraic uncertainty estimates from the two solutions by more than 10^6. Those estimates use the actual observation weights, residuals and smallest eigenvalues; they are not certified bounds. The trend supports reduced mesh dependence of these displacements, not a known continuum error.

6.2 Solver behaviour and conditioning

Section 4's recorded timings show why iteration count alone is not a speed ranking. Diagonal PCG used 226 steps against CG's 242, but their minimum setup-inclusive times were 4.99 and 4.31 ms, respectively. Full GMRES used 228 steps yet took 146 ms. These imported measurements follow Section 3's protocol; no solver timings were repeated here, and small differences are not evidence of a general ranking.

Meanwhile, conditioning worsens under refinement despite the settling displacements. At $n = 20$, symmetric diagonal scaling $P^{-1/2}AP^{-1/2}$ reduces the estimated condition number from 9104 to 8029, where $P = \mathrm{diag}(A)$ as in Section 3. This modest improvement does not guarantee faster PCG. The spectral diagnostics also explain the failures. Ordinary Jacobi's iteration matrix has spectral radius about 1.96, meaning that some error components grow. Damping by 0.5 reduces this to about 0.999816, below one but close enough to predict slow asymptotic decay [2]. This agrees with the recorded divergence of Jacobi and budget exhaustion of damped Jacobi. Steepest descent also exhausted the budget. Their runtimes are not times to the requested accuracy. Among accepted runs, the largest relative difference from the same-mesh LU reference was 1.74×10^{-7}, for Gauss–Seidel, despite its residual being below 10^{-8}. Residual and displacement error are different quantities.

6.3 Sensitivity to small load changes

At the fixed 20-cell mesh, we compare two load changes of equal relative size, $\|\delta b\|_2/\|b\|_2 = 10^{-3}$. The first scales the original load. The second redistributes horizontal traction along the right edge, with its increment proportional to $2y/L - 1$, where $L = 1$ m is the edge height. Its integrated added force is zero, but its moment changes. The coefficient is chosen to give the stated load-vector norm; edge forces are integrated consistently.

We solve both changed systems with the same matrix and measure

$$G = \frac{\|\delta x\|_2/\|x\|_2}{\|\delta b\|_2/\|b\|_2}, \tag{1}$$

where x is the baseline free-displacement vector and δx its change. Uniform scaling gives $G \approx 1$, as linearity predicts, while redistribution gives $G \approx 1.77$. Both are far below $\kappa_2(A) \approx 9104$: conditioning describes possible amplification, not what every load direction produces. Neither test searches for the worst case. The Euclidean norm here compares free nodal components on one fixed mesh, not displacement fields on different meshes.

6.4 Validation and remaining limits

Across the four refinement references, prescribed displacements are exactly zero and true relative residuals are below 1.3×10^{-13}. Normalized force and moment imbalances are below 4×10^{-14}, counting reactions only at constrained components. Force imbalance is normalized by the sum of applied nodal force magnitudes; moment imbalance about the origin uses that force scale times the square's diagonal length. The companion data retain each check separately. The separate roller-supported affine test gives a relative nodal displacement error of 2.12×10^{-15} against its exact field; it does not provide an exact answer for the clamped benchmark.

The evidence therefore separates algebraic solution accuracy, mesh dependence and verification against a known solution. It does not yet establish continuum accuracy or physical validity near clamp corners. The rectangle, perforated square, load-patch and shortened-clamp studies remain unexecuted, and no material or thickness comparison is claimed. Their interpretation requires checked reference solutions and preserved physical loads, not merely similar-looking deformation plots. The assembly-routine permission noted in Section 2 and final case selection also remain unresolved.

References

[1] NLA project team, *NLA: force-driven square and solver benchmark*, `https://github.com/Pueblo98/NLA`, 2026, Source snapshot 9736c8e; unchanged implementation files accompany the section's reproducibility package.

[2] Yousef Saad, *Iterative methods for sparse linear systems*, second ed., SIAM, 2003.